

Product Requirements Document

Interactive Travel Planning & Documentation Assistant

Document Version:	1.0
Date:	May 5, 2026
Project Name:	Travel Planner Web Application
Tech Stack:	Next.js, shadcn/ui, D2, Travel APIs
Author:	Greesh Varthan

1. Product Summary

The website is a travel planning assistant that guides users through a question-based flow to build a complete trip plan. It asks about destination, dates, budget, passport nationality, visa needs, travel documents, and trip preferences, then generates an itinerary, checklist, and map-based trip view. The product should feel interactive and responsive, not like a static form, and should help users understand documentation requirements before booking.

2. Product Goals

The primary goal is to reduce the time and confusion involved in planning a trip by combining trip research, document prep, and booking organization in one place. Secondary goals are to help users determine whether they need a visa, what documents they should prepare, and which flights and hotels best match their budget. The website should also provide official destination links and a visual flowchart so users can understand the trip-planning path.

3. Target Users

The main users are solo travelers, couples, families, students, and budget-conscious travelers planning domestic or international trips. A second audience is travelers who need help with documentation, such as passport validity, visa checks, medical paperwork, or child-travel requirements. A third audience is users who want a centralized travel dashboard instead of browsing multiple websites for flights, hotels, maps, and official guidance.

4. Core User Experience

The experience begins with a guided questionnaire that behaves like a flowchart. As the user answers each question, the app updates the checklist, travel requirements, and suggested bookings in real time. The user should be able to switch between itinerary, checklist, map, and document views without losing progress, and the layout should make the trip status easy to understand at a glance.

5. Functional Requirements

5.1 Trip Questionnaire

The app must ask users for trip basics: origin, destination, travel dates, budget, and number of travelers. It must also ask documentation-related questions such as passport country, passport expiration date, visa or travel authorization need, travel with children, medication needs, and whether the user plans to drive abroad. Answers should drive downstream logic for the checklist and itinerary.

5.2 Visa and Documentation Logic

The app must determine whether the traveler needs a visa, entry authorization, or other document preparation steps based on destination and passport country. It must link to official government or embassy pages for the latest entry rules. The system should flag incomplete or risky document situations, such as soon-to-expire passports or missing travel paperwork.

5.3 Budget-Based Planning

The app must generate trip suggestions that respect the user's budget. It should rank flight and hotel options by total cost, convenience, and distance to the destination or city center. If no options fit the budget, the app should suggest tradeoffs such as shifting hotel location, changing flight timing, or lowering trip length.

5.4 Itinerary Builder

The app must create a structured itinerary from the user's selected bookings and preferences. It should support day-by-day entries, travel times, hotel check-in and check-out, and editable activities. The itinerary should remain updateable after initial generation so users can refine plans without rebuilding from scratch.

5.5 Interactive Map

The app must show an interactive map with the airport, hotel, planned route, and nearby points of interest. The map should display distances and approximate travel times when possible. The map view should stay connected to the itinerary so users can understand how hotel location affects their schedule.

5.6 Checklist Generation

The app must generate a destination-specific checklist based on the user's answers. Checklist items should include passport validity, visa, travel insurance, booking confirmations, medication documents, child-travel documents, and any driving-related requirements. Users should be able to mark items complete and see checklist progress.

5.7 Official Links

The app must automatically provide links to official destination resources, such as government travel pages, visa pages, and travel advisories. These links should be surfaced inside the checklist or documentation section so users can verify requirements directly from authoritative sources.

6. Information Architecture

The website should include these main pages: Home, Trip Questionnaire, Results Dashboard, Itinerary, Map, Checklist, Documents, and Saved Trips. The homepage should be simple and action-oriented, while the dashboard should unify the trip summary, flowchart, and checklist. The checklist and itinerary should be visible as live outputs of the questionnaire rather than separate disconnected tools.

7. UI and Interaction Requirements

The frontend should use shadcn/ui components for the main interface structure, including cards, tabs, accordions, sheets, dialogs, and progress indicators. The design should feel modern, modular, and responsive, with clear state changes as the user answers questions. D2 should be used to render the flowchart that shows how the questionnaire branches into different travel requirements.

8. Flowchart Behavior

The flowchart should represent the trip logic visually and update as the user completes the questionnaire. It should begin with trip type, then branch into documentation checks, budget planning, bookings, and final trip assembly. Users should be able to see where they are in the process and which questions are still pending, making the planner feel like a guided decision system.

9. Data Model

The app should store: User profile, Trip profile, Passport country and expiration, Visa status, Flight options, Hotel options, Booking confirmations, Itinerary entries, Checklist items, and Official source links. This structure supports both planning and document verification, while keeping the data reusable for future trips. It also matches the kind of stored trip and document organization users expect from a travel planning platform.

10. Suggested Tech Stack

Component	Technology	Purpose
Frontend Framework	Next.js	Server-side rendering and routing
UI Components	shadcn/ui	Interactive, accessible UI elements
Diagram Generation	D2	Flowchart and decision tree visualization
Maps	Google Maps / Mapbox	Route and location visualization
Travel APIs	Flight & Hotel APIs	Booking search and pricing
Backend	Node.js / Python	API logic and data handling
Database	PostgreSQL / MongoDB	User trips and itinerary storage

11. MVP Scope

The minimum viable product should include: A guided trip questionnaire, Visa/documentation logic, Budget filtering, Flight and hotel suggestions, Auto-generated checklist, Interactive map, D2 flowchart, and Official links section. This is enough to validate the concept without overbuilding advanced features too early. A solid MVP can already deliver the core value of reducing travel planning friction and documentation uncertainty.

12. Success Criteria

The product is successful if users can complete a trip plan in one session, understand their document requirements clearly, and save time compared with manual planning. Additional success signals are higher checklist completion, saved trips, and repeated use for future travel. A strong indicator of product fit will be whether users rely on the app for both booking organization and pre-trip documentation guidance.